Taeyoung Yeon taeyoungyeon@nyu.edu • taeyoungyeon.github.io

I build practical real-time human sensing applications for consumer devices. At Northwestern, I focused on enabling smartwatches, AR glasses, and XR headsets to sense pose, body trajectory, and activity on-device, using sensors already built in (e.g., IMUs, microphones). In my PhD, I am expanding these approaches in three directions: context-aware interaction with AI agents, just-in-time interventions for diverse real-world applications, and novel health sensing methods on consumer devices.

Research Interests

Human-Computer Interaction, Practical Human Sensing on Consumer Devices, Interaction with AI Agents, On-Device Machine Learning

Education

New York University, Tandon School of Engineering September 2026 – Present
Ph.D. Student, Human-Centered Technology, Innovation & Design *Brooklyn, NY*
Advisor: Prof. Vedant Das Swain

Seoul National University March 2017 – February 2024
Bachelor of Science in Computer Science and Engineering *Seoul, South Korea*
Overall GPA: 3.53/4.00, *Graduated **cum laude***

Research Experience

Research Assistant September 2026 – Present
New York University, COPHEE Lab, advised by Prof. Vedant Das Swain *Brooklyn, NY*

- Working on cognitive load and context-aware AI agents with wearable sensing.

Post-Baccalaureate Research Fellow September 2024 – June 2026
Northwestern University, SPICE Lab, advised by Prof. Karan Ahuja *Evanston, IL*

- Contributed to 6 research papers (3 published, 3 under review) spanning on-device ML, human activity sensing, and real-time XR avatars.

Publications

** denotes equal contribution*

Conference Papers

[C1] Chenfeng Gao*, **Taeyoung Yeon***, Sungheon Park, Vasco Xu, Anish Prabhu, Mar Gonzalez-Franco, Karan Ahuja. **FlowAvatar: Real-Time Full-Body Avatars from Sparse Egocentric Inputs on Consumer XR Devices.** *To appear in Proceedings of the 2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR '26).* 2026.

[C2] Yiquan Li*, **Taeyoung Yeon***, Chenfeng Gao, Vasco Xu, Xuanyou Liu, Karan Ahuja. **MARIO: Motion-Augmented Real-Time Multi-Sensor Inertial Odometry.** *Proceedings of the 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition, Findings (CVPR '26).* 2026.

[C3] **Taeyoung Yeon**, Vasco Xu, Henry Hoffmann, Karan Ahuja. **WatchHAR: Real-time On-device Human Activity Recognition System for Smartwatches.** *Proceedings of the 27th ACM International Conference on Multimodal Interaction (ICMI '25)*, pp. 387–394. 2025.
DOI: 10.1145/3716553.3750775

Under Review

[U1] **Taeyoung Yeon** (first author). Manuscript under review at the 2027 IEEE International Conference on Robotics and Automation (ICRA '27).

[U2] Co-authored manuscript under review at the 2027 ACM CHI Conference on Human Factors in Computing Systems (CHI '27).

[U3] Co-authored manuscript under review at the 2027 ACM CHI Conference on Human Factors in Computing Systems (CHI '27).

Professional Experience

Full Stack Game Developer (Alternative Military Service) September 2019 – February 2023
Awesomepiece *Seoul, South Korea*

CUDA Engineer Intern July 2019 – August 2019
MindsLab *Pangyo, South Korea*

Research Demonstrations

Presented research demonstrations to industry partners: **Google Platform and Experiences, Apple, Qualcomm XR, Adobe Research, Dolby Studios**.

Honors & Awards

School of Engineering PhD Fellowship, NYU Tandon School of Engineering 2026 – 2027

Outstanding Reviewer Recognition (2), UbiComp/IMWUT 2026

Academic Services

Reviewer: CHI 2027, UbiComp/IMWUT 2026, ICMI 2026

Technical Skills

Programming Languages: Python, Java, C#, JavaScript, Kotlin, Swift, CUDA

Machine Learning: TensorFlow, PyTorch, CoreML, scikit-learn, model optimization

Mobile & VR Development: Unity (Meta SDK), iOS/Android development, cross-platform frameworks

Backend & Infrastructure: Spring Boot, Netty, Django, React, MySQL, Google Cloud Platform

References

Prof. Vedant Das Swain
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Department of Technology Management and Innovation
NYU Tandon School of Engineering
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Prof. Karan Ahuja
Lisa Wissner-Slivka & Benjamin Slivka
Assistant Professor, Department of Computer Science
Northwestern University
kahuja@northwestern.edu